MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, RIGHT ANGLE, 90 THRU 240 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification sheet is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

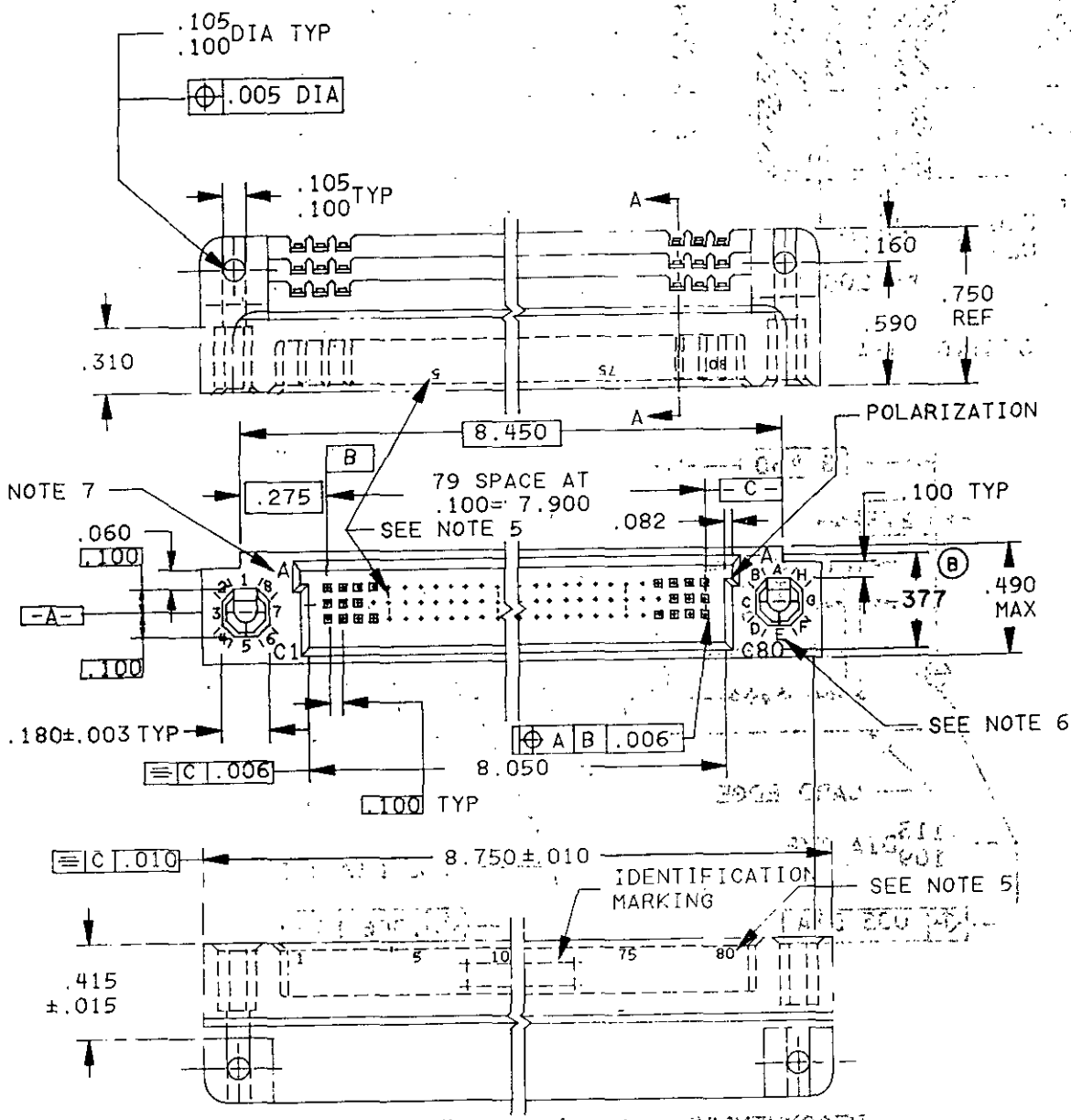
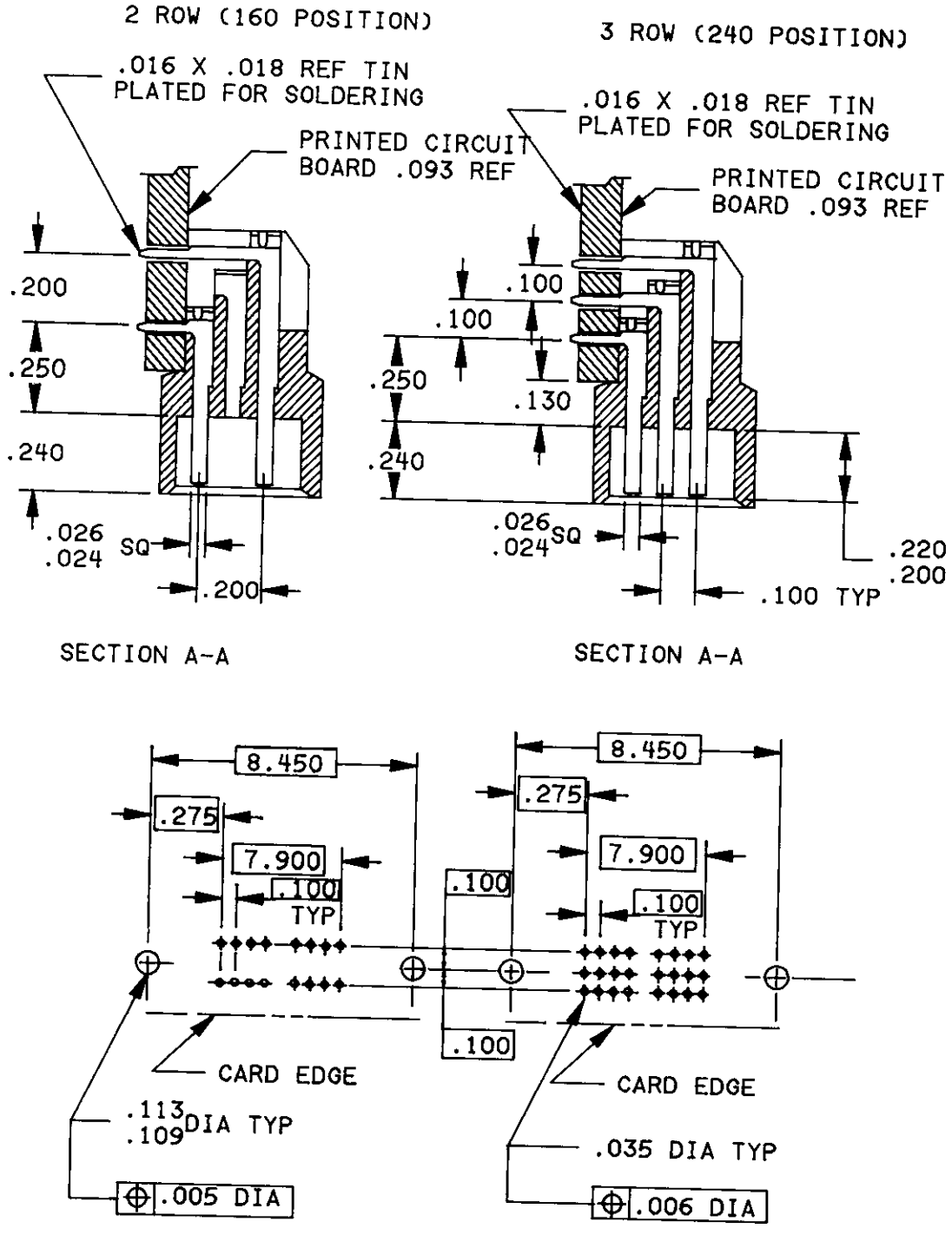


FIGURE 1. Connectors, kit plug (.100 spacing) 160 and 240 contacts

(B) Denotes change

FSC 5935

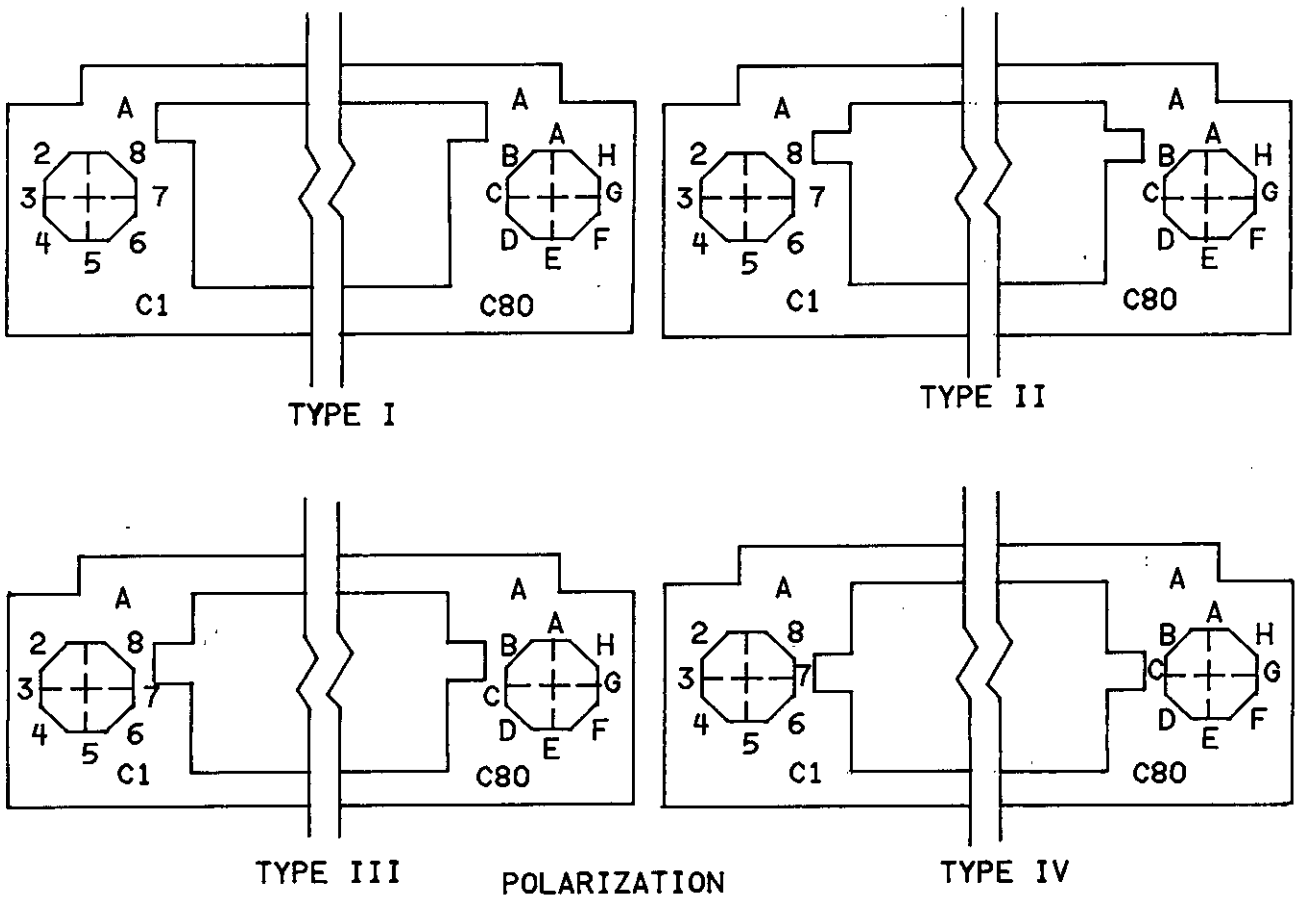
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RECOMMENDED CIRCUIT LAYOUTS

FIGURE 1. Connectors, kit plug (.100 spacing) 160 and 240 contacts - Continued.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals and +2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/130.
5. Numbers indicating end cavities stamped both sides. Lines indicating every five cavities stamped on mating face, except end cavities.
6. Location indicators embossed on surface.
7. Letters indicating rows and numbers indicating end cavities stamped this surface.

INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.109	2.76	.310	7.87
.005	.13	.113	2.87	.377	9.58
.006	.15	.130	3.30	.415	10.54
.010	.25	.160	4.06	.490	12.54
.024	.64	.180	4.57	.590	14.99
.026	.66	.200	5.08	.750	19.05
.035	.89	.220	5.59	7.900	200.66
.060	1.52	.240	6.10	8.044	204.32
.082	2.08	.250	6.35	8.450	214.63
.100	2.54	.275	6.99	8.750	222.25
.105	2.67				

FIGURE 1. Connectors, kit plug (.100 spacing) 160 and 240 contacts - Continued.

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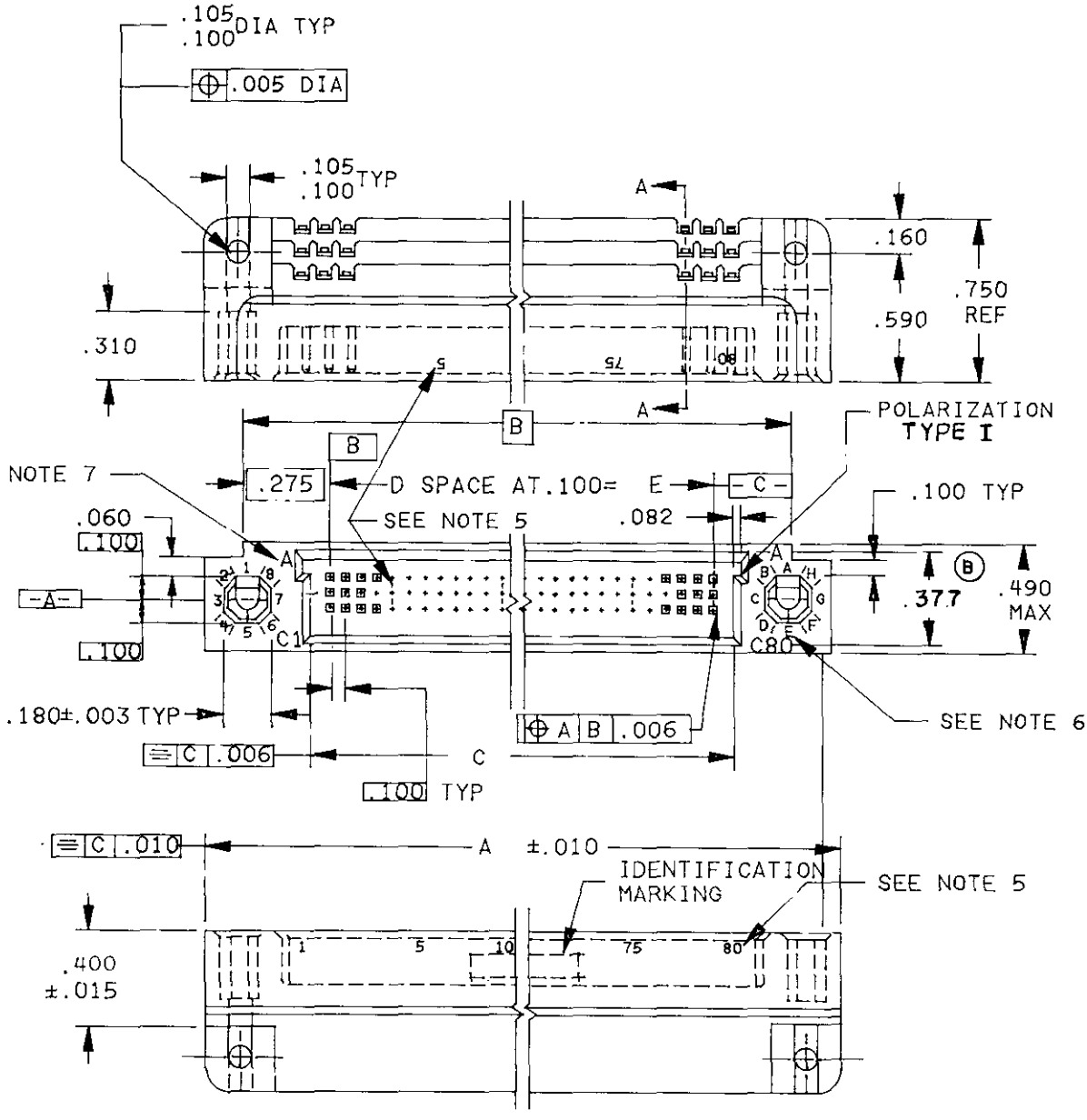
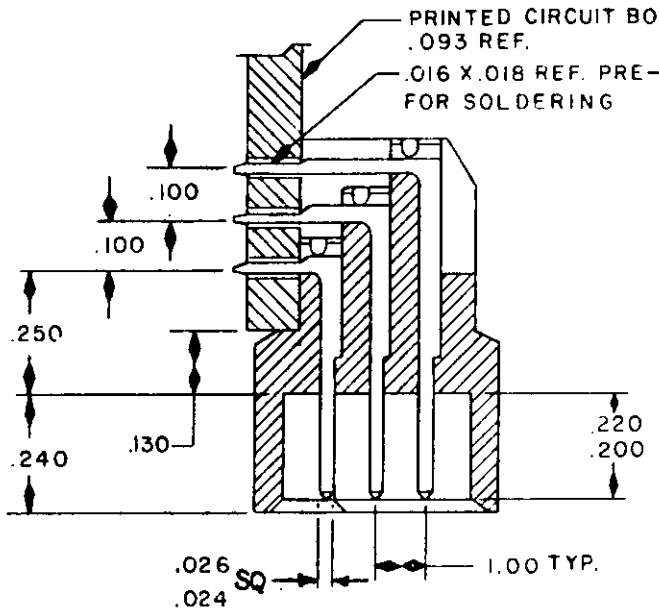
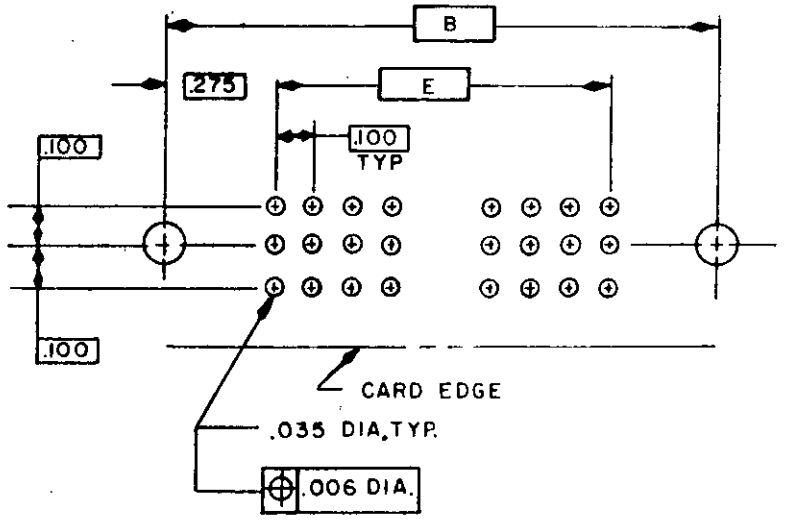


FIGURE 2. Connectors, plug (.100 spacing) 90 thru 240 contact positions

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SECTION A-A



INCHES	MM
.003	.08
.005	.13
.006	.15
.010	.25
.024	.64
.026	.66
.035	.89
.060	1.52
.082	2.08
.100	2.54
.105	2.67
.109	2.76
.113	2.87
.130	3.30
.160	4.06
.180	4.57
.200	5.08
.220	5.59
.240	6.10
.250	6.35
.275	6.99
.310	7.87
.377	9.58
.415	10.54
.490	12.54
.590	14.99
.750	19.05
7.900	200.66
8.044	204.32
8.450	214.63
8.750	222.25

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals and +2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/130.
5. Numbers indicating end cavities stamped both sides. Lines indicating every five cavities stamped on mating face, except end cavities.
6. Location indicators embossed on surface.
7. Letters indicating rows and numbers indicating end cavities stamped this surface.

FIGURE 2. Connectors, plug (.100 spacing) 90 thru 240 contact positions - Continued.

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TABLE I. Dash numbers, polarizatioon type, and contact type for 160 and 240 contact position.

Dash number	Number contacts	Type polarization
01	160	I
02	160	II
03	160	III
04	160	IV
05	240	I
06	240	II
07	240	III
08	240	IV

Ⓑ TABLE II. Dash numbers and dimensions. (For 90 thru 240 contact positions)

Dash no.	No. of contacts	A	B	C	D	E	Contact ident. no.
09	90	3.750 (95.25)	3.450 (87.63)	3.044 (77.32)	29	2.900 (73.66)	1-30
10	105	4.250 (107.95)	3.950 (100.33)	3.544 (90.02)	34	3.400 (86.36)	1-35
11	120	4.750 (120.65)	4.450 (113.03)	4.044 (102.72)	39	3.900 (99.06)	1-40
12	135	5.250 (133.35)	4.950 (125.73)	4.544 (115.42)	44	4.400 (111.76)	1-45
13	150	5.750 (146.05)	5.450 (138.43)	5.044 (128.12)	49	4.900 (124.46)	1-50
14	165	6.250 (158.75)	5.950 (151.13)	5.544 (140.82)	54	5.400 (137.16)	1-55
15	180	6.750 (171.45)	6.450 (163.83)	6.044 (153.52)	59	5.900 (149.86)	1-60
16	195	7.250 (184.15)	6.950 (176.53)	6.544 (166.22)	64	6.400 (162.56)	1-65
17	210	7.750 (196.85)	7.450 (189.23)	7.044 (178.92)	69	6.900 (175.26)	1-70
18	240	8.750 (222.25)	8.450 (214.63)	8.044 (204.32)	79	7.900 (200.66)	1-80

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1, 2, and table II.

Material: In accordance with MIL-C-55302, except body material type GPT-30F per MIL-M-24519.

Plating:

Contact engagement area (0.220 inch minimum length): Gold (per MIL-G-45204, type II, grade C, class 1) over nickel (per QQ-N-290, class 2, 50 to 150 microinches).

Solder tail area (0.160 inch minimum length): Tin lead plate (per MIL-T-10727, type I, 100 microinches minimum) over nickel (QQ-N-290, class 2, 50 to 150 microinches).

Contact identification: See figures 1 and 2.

Current rating: 3.0 amperes, maximum per contact: 2.25 amperes continuous per connector at room ambient.

Keying: See MIL-C-55302/31. Two keys and two key retaining rivets packaged with connectors for -01 through -08 only. For -09 through -18, keys and rivets shall be ordered separately.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by 0.25; the withdrawal force in pounds shall be a minimum of 0.025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding 0.020 ohm.

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level: 900 volts rms.
High altitude: 200 volts rms.

Part number: M55302/129-(and dash number from table I for 160 and 240 contact positions).

M55302/129-(and dash numbers from table II for 90 through 240 contact positions).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-55302/129B(EC)	CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, RIGHT ANGLE, 90 THRU 240 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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